

lecture 1

24-09-2024

Theory of Automata

Thought process
of AI / computer
science.

↓
greek word (something that
works automatically)

Machines
& Models

DFA, NFA, Moore / Mealy Machine,
PDA, TM

→ Language + Machine.
↓
(logical structure)

(set
of alphabets
based
on words)

(model the
human thought
process)

$\Sigma \rightarrow \text{sigma}$
(set of alphabets)

Example of language..

$L =$ All words that start with 'a' over

$\Sigma = \{a, b\}$

↓
examples-

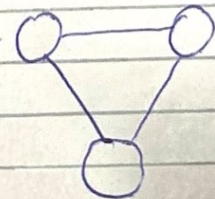
$\{a, aa, ab, abb, aab, \dots\}$

↓
['a' se start
hon or bs
a or b he
use hon.]

→ regular expression

↳ formal way to define
a language.

→ graphs



a sort of a tree
graph (represent a
machine)

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Objective

- fundamental capabilities and Limitations of the computers.
- express and solve problems.
- problem solving.
- designing new programming language, compilers, string searching etc.
- mathematical problem.
 $-((2+3) + (10-3) / 10)$

Example.

All words ends at 'b'

$$\Sigma = \{ a, b, c \}$$

$\{ b, bb, ab, bab, bbb, aabbab, cb, \dots \}$

→ set of all alphabet.
(only using a, b, c)

ddacb → invalid.

- collection of all successful valid input is called language.

Automata Theory

- Abstract machine → Automata
- Automata is greek letter.
- Automata is a word that means something that works automatically. It is plural of term automation.

"SET Theory

SETS

Def:- A set is collection of are:-

- unordered
- distinct
- same to
- with or

- * order donot it matters
- * position donot
 $\{a\} \rightarrow \text{set}$
 $a \rightarrow \text{element}$

① $X \cdot n :: n^2$

$$X=1 = 1^2$$

$$X=2 = 2^2$$

②

③

$$2n + 1 ::$$

$$2(0) + 1 ::$$

$$2(2) +$$

$$2(3) +$$

2024

"SET Theory"

(Lecture 2)

SETS

Limitation

Def: A set is well defined collection of objects which are:-

- unordered
- distinct
- same type
- with common properties.

- * order donot matter in sets $\{\}$ but it matters in sequence $\langle \rangle$.
- * position donot matter in set.
 - $\{a\} \rightarrow \text{set}$
 - $a \rightarrow \text{element}$

cb, cb --- }

① $x \cdot n :: n = 1, 2, 3, \dots$

$$x \cdot 1 = 1x = \{x, 2, 4, 6, \dots\}$$

$$x \cdot 2 = 2x = \{x, x, \dots\}$$

②

$$2n :: n = 1, 2, 3$$

$$2(n) = 2(1) = 2$$

$$2(2) = 4$$

$$2(3) = 6$$

valid

③

$$2n + 1 :: n = 0, 1, 2, \dots$$

$$2(0) + 1 = 1$$

$$2(2) + 1 = 5$$

$$2(3) + 1 = 7$$

on.

4) $C = \{a, b, c, d, e, f, g, h, i, j, k\}$

$C = \{a, b, \dots, k\}$ finite

Union ~~(common)~~ All

$A \cup B$ (All elements in A & B)

Intersection ~~(all)~~ (common)

$A \cap B$ (Common elements in A and B)

Difference

$A - B$

$B - A$

complement

$U = \{1, 2, \dots, 10\}$

$A = \{1, 2, 3\}$

$\overline{A} = \{4, 5, 6, 7, 8, 9, 10\}$

$\overline{\overline{A}} = A$

$\overline{A} = U - A$

Demorgan's law.

$\overline{A \cap B} = \overline{A} \cup \overline{B}$

1-09-2024

Empty, Null set ϕ

$$\phi = \{ \}$$

$$S \cup \phi = S$$

$$S \cap \phi = \phi$$

$$S - \phi = S$$

$$\phi - S = \phi$$

↓
any set.

$\bar{\phi}$ = Universal set.

A & B)

in

$$\bar{\phi} = U - \phi = U$$

Subset

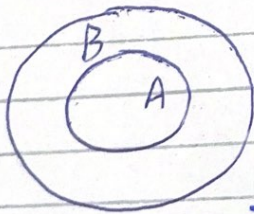
$$A = \{1, 2, 3, 5\}$$

$$B = \{1, 2, 3, 4, 5\}$$

$$A \subset B$$

Proper subset

$$A \subset B$$



Disjoint set

$$A \not\subset B$$

no element is
present in other
set.

Cardinality $\rightarrow$ size / no. of element.

$\rightarrow$ for finite sets

$$A = \{2, 5, 7, 1, 10\}$$

$$|A| = 5$$

Power set

$\rightarrow$ set of all subset of a set.

$$S = \{a, b, c\}$$

$$2^S = \{ \emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\} \}$$

$$\text{observation: } |2^S| = 2^{|S|}$$

$$8 = 2^3$$

Cartesian Product

$$A = \{2, 4\}$$

$$B = \{2, 3, 5\}$$

$$C = \{7, 8\}$$

$$A \times B = \{ (2, 2), (2, 3), (2, 5), (4, 2), (4, 3), (4, 5) \}$$

$$|A \times B| = |A| |B|$$

"language"

lecture 3.

→ certain combination of letters, words, alphabets that lead to a sentence.

All words whose length is greater than equal to 2 and are made of combination of x and y .

$$\Sigma = \{x, y\} \\ = \{x, y, yx, xx, yy, xxy, xyy, \dots\}$$

sigma $\Sigma \rightarrow$ set of Alphabets
Ta $\& T \rightarrow$ set of words

- **words** are meaningful
egs- like $\rightarrow$ meaningful

- **string** $\rightarrow$ meaningless
egs: ekli $\rightarrow$ meaningless

- Every word is a string but every string is not a word.

Representation of length of a word.
 $|W| = \text{all}$

$|W|_2 3 \rightarrow$ length
or

$W = 011$

$|W| = 3$

finite language
e.g. English words

infinite language.
e.g. English sentences

De

→ define
→ define
and

(finite)
All words whose length is exactly 2
over the sigma 0,1
 $\Sigma = \{0,1\}$

→ C
→
→
→

$\Sigma = \{00, 11, 01, 10\}$

1, ϵ , $\lambda \rightarrow$ empty / Null string.
whose length is less than 3

E

$\Sigma = \{\lambda, 0, 1, 00, 11, 01, 10\}$
↓
empty string

e

How to check if alphabet belongs to a
finite $\rightarrow$ check in Ta
infinite $\rightarrow$ check ~~if~~ according
to the rule of
language.
↓
a language
or not.

Defining language.

- define alphabet set.
- define rules for forming valid words and sequence of words from Σ .
- Called grammar
- Can be descriptive
- Can be mathematical.
- Can also define supporting functions

Example = $\Sigma = \{a, b, \dots, z\}$

Palindrome = $\{ \wedge, \text{all string } x \text{ such that reverse}(x) = x \}$

eg: mom

→ $L = \{ \text{all words formed only of odd number of } x \}$

$$\Sigma = \{a, b\}$$

1) All words whose length is even

$$f = \{ aa, bb, aaaa, bbbb, aabb, bbaa, aaabbb, bbbbaa, \dots \}$$

2) All words whose second last symbol is 'b'

$$f = \{ ba, aaba, aabb, baba, abba, \dots \}$$

3) All words that do not start with double letter.

$$f = \{ ab, ba, baba, abba, aba, \dots \}$$

↓
1, a, b
↓
Empty string

Kleene Closure

↳ Denoted by Σ^*

A set of all the strings (finite) that can be formed by the elements of Σ (set of all alphabets) where elements may be repeated any number of times.

string $\rightarrow$ finite

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{ \epsilon, a, b, ab, ba, aa, bb, aaa, bbb, aba, bab, \dots, aaaa, bbbb, abab, baba, \dots, aaab, baab, \dots, aaaa, bbbb, abbb, baab, \dots \}$$

Kleene closure

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \dots$$

plus operation

$$\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \dots$$

$\hookrightarrow$ do not contain empty string.

example:

$$\Sigma = \{a, b\}$$

$(\epsilon) \times$

$$\Sigma^0 = \{ \epsilon \} \rightarrow \text{empty set}$$

$$\Sigma^1 = \{a, b\}$$

$$\Sigma^2 = \{ \{a, b\}, \{a, b\} \}$$

(Cartesian product) $\Sigma^2 = \{aa, ab, ba, bb\}$

$$\Sigma^3 = \{ \{a, b\}, \{a, b\}, \{a, b\} \}$$

$\vdots$

Σ^+ do not contain null/empty string or Σ^0 .

→ If $\Sigma = \phi$ then

$$\rightarrow \Sigma^* = \{ \Lambda \}$$

→ $\Sigma^* = (\Sigma^*)^*$ so on.

Example:-

$$a^* = \{ \Lambda, a, aa, aaa, aaaa, \dots \}$$

$$(ab)^* = \{ \Lambda, ab, abab, ababab, \dots \}$$

$$(a^*)^* = (a^*)^*$$

agr koi string phley he infinite ha or us ka loop to to wo infinite (same) he rahey gi.

→ concatenation of string.

$$w = a_1 a_2 \dots a_n \quad b_1 b_2 \dots b_m$$

$$w = abb$$

$$x = bbab$$

$$wx = \overline{abb} \overline{bbab}$$

$$xw = \overline{bbab} \overline{abb}$$

length of string

$$|aabb| = |4|$$

Reverse of string

$$w^R = a_n \dots a_2 a_1$$

$$\begin{array}{l} w = abb \\ w^R = bba \end{array} \quad \rightarrow$$

length of concatenated string

$$w = abb$$

$$x = bab$$

$$w + x = |3 + 4| = 7$$

Empty string

$$|\lambda| = 0$$

$$w = abb$$

$$\lambda w = w \lambda = 3$$

$$\begin{array}{cc} \downarrow & \downarrow \\ 3 & 0 \end{array}$$

substring

string

abbab

abbab

abbab

substring

ab

abba

bab

Example:-

$$\Sigma = \{0, 1\}$$

all string that contain ~~01~~ 01
as a substring.

$$f = \{ 01, 010011, 001100, 101001, \\ 0101001011 \dots \}$$

OR.

$$f = \{ 01, 001, 101, 0101, 0001, 1001, \\ 001001, \dots \}$$

Prefix and suffix.

$$w = ux$$

prefix $\rightarrow$ suffix

$\wedge 101 \wedge$

prefix $\Rightarrow$ suffix

$$\wedge 101 \Rightarrow 101$$

$$1 \rightarrow 01$$

$$10 \rightarrow 1$$

$$101 \rightarrow \wedge$$

(A) sb key
sath aa
skta
ha.

power

$$w = aab$$

$$w = ww$$

~~aaab~~

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Prefixes

λ
a
ab
abb
abba
abbab

Suffixes

abab
babab
bab
ab
b
 λ

$w \in u^*$

↳ suffix
prefix

Σ^* → All possible set of alphabet
is Kleene closure Σ^* .

$$\Sigma = \{a, b, c\}$$

$$\Sigma^* = \{ \lambda, a, b, c, ab, ac, aa, aaa, aba, \dots \}$$

→ Language is any subset of Σ^*

→ set $\Phi = \{ \} \neq \{ \lambda \}$
→ set size $|\{ \}| = |\Phi| = 0$

→ set size $\neq |\{ \lambda \}| = 1$

→ string length $|\lambda| = 0$

↳ empty set string length is zero.

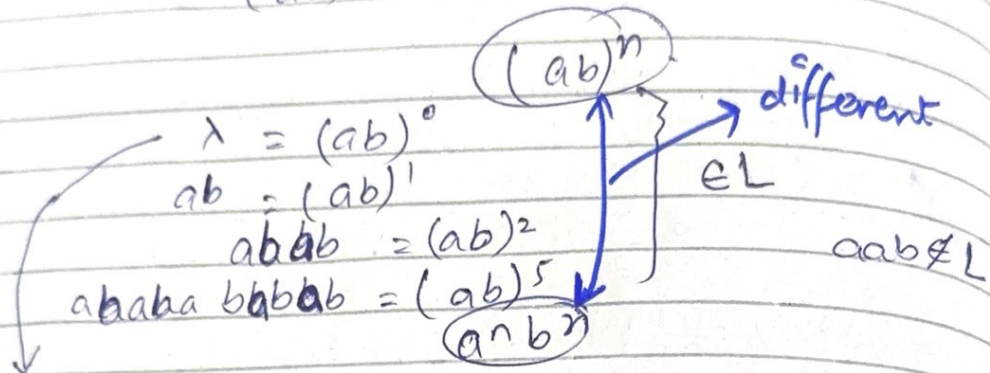
empty set

is present in a set so
length is 1.

Infinite Language

$$L = \{a^n b^n : n \geq 0\}$$

$$\begin{aligned}
 w^2 &= ww \\
 (ab)^2 &= abab \\
 (ab)^0 &= \lambda \\
 (ab)^3 &= ababab
 \end{aligned}$$



$$\begin{aligned}
 a^0 b^0 &= \lambda \\
 a^1 b^1 &= ab \\
 a^2 b^2 &= aabb \\
 a^5 b^5 &= aaaaa bbbbbb
 \end{aligned}$$

$n = \text{same}$. for
 a and b
 both.

$ba \Rightarrow$ not
 possible.

→ phley a aye ga
 phir b aye ga.

Operations on Language.

$$\{a, ab, aaaa\} \cup \{bb, ab\} = \{a, ab, bb, aaaa\}$$

Example:

$$\Sigma = \{a, b\}$$

$$L = \{a, ab, aa, aab, aaa, aba, \dots\}$$

$$\bar{L} = \Sigma^* - L \quad (\text{complement})$$

→ universal set.

$$\Sigma^* = \{\lambda, a, b, ab, ba, bb, aa, aaa, bbb, aba, \dots\}$$

$$L = \{ \Lambda, b, ba, bb, bab, baa, \dots \}$$

Reverse of language

$$L^R = \{ w^R : w \in L \}$$

$$w = abb$$

$$w^R = bba$$

$$L = \{ ab, baa, bba \}$$

$$L^R = \{ ba, aab, abb \}$$

concatenation of language

$$L_1 L_2 = \{ xy : x \in L_1, y \in L_2 \}$$

$$L_1 = \{ ab, bb, abb \}$$

$$L_2 = \{ 00, 11 \}$$

$$L_1 L_2 = \{ ab, bb, abb \} \{ 00, 11 \}$$

$$= \{ ab00, ab11, abb00, abb11 \}$$

Another operation

$$L^n = \underbrace{LL \dots L}_n$$

$$\{a, b\}^3 = \{a, b\} \{a, b\} \{a, b\}$$

special case.

$$L^0 = \{\lambda\} \cup \{a, bba, aab\}^0 = \{\lambda\}$$

$$L_1 = \{ab, bb, abb\}$$

$$L^0 = \{\lambda\}$$

$$L_1' = \{ab, bb, abb\}$$

$$L_1 L_1 = \{ab, bb, abb\} \{ab, bb, abb\}$$

$$= \{abab, abbb, ababb, abbab, abbb, abbab\}$$

Star-closure (Kleene *)

$$L^* = L^0 \cup L^1 \cup L^2 \dots$$

Example:

$$L = \{a, bb\}$$

$$L^0 = \{\lambda\}$$

$$L^1 = \{a, bb\}$$

$$L^2 = \{a, bb\} \{a, bb\}$$

$$L^* = \{\lambda\} \cup \{a, bb\} \cup \{aa, abb, bba, bbbb\}$$

$$L^* = \{\lambda, a, bb, aa, abb, bba, bbbb\}$$

Positive-closure

$$L^+ = L^1 \cup L^2 \cup L^3 \dots$$

$$= L^+ - \{ \lambda \}$$

$$\{a, bb\}^+ = \left\{ \begin{array}{l} a, bb \\ aa, abb, bba, bbbb, \\ aaa, aabb, abba, abbbb, \dots \end{array} \right\}$$

Recursive language Def

↳ Characteristically a three step process.

step 1: 2 is in P-Even

↓
positive-even number.

step 3:

$$x+2 = 2+2 = 4$$

$$x+2 = 4+2 = 6$$

$$x+2 = 6+2 = 8$$

$$x+2 = 8+2 = 10$$

step 2: if x is positive even then $x+2$ is also positive even.

$$\Sigma = \{a, b\} \rightarrow \text{zero length}$$

$$\text{palindrome} = \{ \lambda, a, b, aa, bb, aba, aab, bab, bba, aaaa, bbbb, abba, \dots \}$$

~~recursive def:~~
palindrom

step 2.. If x is palindrome then
 axa and bxb are
also palindrome.

step 3.. only those are constructed
with step 1 and step 2.

if axa $x = \lambda$
 $a\lambda a$ palindrome.

aaa $x = a$ palindrome
 aba $x = b$ palindrome.

if bxb $x = \lambda$
 $b\lambda b$ palindrome.

bab $x = a$ } palindrome
 bbb $x = b$

if axa $x = aba$
 $aabaa$ palindrome

;
so on.

10-10-2024

Arithmetic Expression by recursive Method:

→ Step 1: Any number (+ve or -ve and zero) are in Arithmetic Expression.

→ Step 2:

If x is AE then
 (x)
 $(-x)$

if x and y are in AE,
 then $(x+y)$, $(x-y)$,
 $x * y$, x / y .

$9 \checkmark$
 $-9 \checkmark$
 $(-(-9)) \checkmark$
 $-(-(-9)) \checkmark$

example:

$\underbrace{[(9+3)-(7*2)]}_{1 \text{ Expression}} + \underbrace{(3+4)}_{2 \text{ expression}}$

Agr do Arithmetic expression han to wo opas mai add, sub, multiply ya divide ho sakti han.

Step 3: only those which are constructed by step 1 and step 2 are correct.

two operators cannot be written together.

$*/, +- , --$ etc ←

Regular Expression.

↳ network security.

⇒ operators:-

1) * ^{power} (star sign).

↳ bold face.

repetition / iteration / loop.
(undefined loop).

always use in power form

a^* , x^* → unknown power.

$a^* = \lambda, a, aa, aaa, aaaa, aaaaa, \dots$

$(ab)^* = \lambda, ab, abab, ababab, \dots$

$ab^* = \underbrace{a}_{=a} \lambda \rightarrow \text{first null.} \therefore (a)(\lambda) = a$
 $ab, abb, abbb, abbbb$

$ab^* = a, ab, abb, abbb, abbbb, \dots$

$a^* b^* = (\lambda)(\lambda) = \lambda$

↳ $\lambda, ab, aabb, aaabbb, aaaaabbbb, \dots$

OR.

$a \times \lambda = a$

$b \times \lambda = b$

$\lambda, a, b, aa, aaa, aaaa, aabb, aaabbb, bbbb, bbbbbb, \dots$

a^*b^* = $(\lambda)(\lambda) = \lambda$
 (All possibilities) $= (a)(\lambda) = a$
 $= (\lambda)(b) = b$
 $= (a)^2(\lambda) = aa$
 $= (a^3)(\lambda) = aaa$
 $= (\lambda)(b)^2 = bb$
 $= (b)^3 = bbb$
 $= (ab) = ab$
 After comma loop begins from start
 $= (a)^2(b)^1 = aabb$
 $= a^3b^3 = aaabbb$
 $= a^1(b)^3 = abbb$
 $\rightarrow \lambda, a, b, aa, aaa, bb, bbb, bbbb, abbb, aabb, aaabbb, \dots$
 (undefined)

2) + (choice)
 (infinite) multiple choice $\rightarrow (a+b) = a, b$
 $\rightarrow$ only one choice
 $\rightarrow (a^* + b) = \lambda, a$
 $(\lambda, b), (a, b), (aa, b), (aaa, b), \dots$
 $\{ \lambda, b, a, aa, aaa, aaaa, \dots \}$
 $\rightarrow (a+b)^*$
 a and b k jitney combination bn skety han
 whole k uper * to null aye ga
 $\lambda, a, b, aa, bb, ab, ba, aaa, bbb, aab, bba, bba, abb, \dots$
 no loop 1 time loop 2 time loop 3 time loop

$(a+b)^+ = \{ \lambda, a, b, aa, bb, ab, ba, aaa, bbb, aab, baa, bba, abb, aaaa, bbbb, abba, aaba, \dots \}$
 $\downarrow$
 All possible combinations of a, b .

$(a+b+c)^+ = \lambda, a, b, c, ca, ab, ca, \dots$
 $\downarrow$
 All possible combinations of a, b, c infinite.

$ab^+ = \{ a, ab, abb, abbb, \dots \}$
 Q

All words that start with 'a' over

$$\Sigma = \{a, b\}$$

$f = \{ a, ab, aa, abb, aab, aaa, abba, \dots abba, abbbba, abbbbbb, aaaaab, \dots \}$

regular expression = $(a^+b^+)^+ a(a+b)^+$
 $\downarrow$
 define only valid words.

All valid words should be formed.